How Models Represent, Store, and Retrieve Knowledge

Module Q13 — DLS 2026

Deep Learning Systems

October 11, 2026

The student's question, sharpened

"How do models represent and store knowledge, and use it to answer?"

- We separate three concerns:
  1. **Representation**: how facts become geometry in activation/weight space.
  2. **Storage**: which parameters hold what, and how retrieval happens.
  3. **Use**: the forward-pass mechanics that turn a prompt into an answer.
- Key tension: knowledge is *distributed* (spread over many weights) yet often *localizable* (editable at specific sites) [7].
- We treat the transformer as a read–write system over a shared residual stream [3].

Representation: the distributional and linear view

- **Distributional hypothesis**: meaning is inferred from co-occurrence; embeddings place similar tokens near one another [1].
- **Linear structure**: relations appear as approximately constant offset vectors (analogies), evidence that attributes are directions, not slots [2].
- **Superposition**: a d-dimensional space stores more features than dimensions by packing them in near-orthogonal directions [4].
- Consequence: a single neuron is usually *polysemantic*; the natural unit of knowledge is a *direction* in the residual stream, recoverable via sparse dictionaries [5].

Storage: feed-forward layers as key–value memories

- Each MLP block computes $\mathrm{FFN}(x) = \sigma(xW_1)\,W_2$.
- Interpret rows of W_1 as **keys** (patterns they fire on) and rows of W_2 as **values** (what they write back) [6].
- Empirically, FFN layers act as a differentiable associative memory: a key matches a context, its value injects a prediction into the residual stream.
- Factual associations are concentrated in mid-layer FFNs at the final token of the subject, enabling causal localization [7].
- Hence "storage" $\approx$ a learned hash of context $\rightarrow$ output contribution.

Use: moving and combining information with attention

- The residual stream is a communication bus; attention *moves* information between positions, FFNs *process* it [3].
- **Induction heads** implement in-context copying/pattern completion and correlate with the emergence of in-context learning [9].
- A simple factual-recall circuit: attention gathers the subject representation $\rightarrow$ mid-layer FFN keys match $\rightarrow$ value writes the object $\rightarrow$ unembedding reads the answer logit.
- **In-context knowledge** (from the prompt) and **parametric knowledge** (from weights) compose in the same stream; conflicts are resolved by relative magnitude of their contributions.

Evidence it is localizable: probing and editing

- **Causal tracing**: corrupt the subject, restore single hidden states, and measure recovery to pinpoint where a fact lives [7].
- **Rank-one editing (ROME)**: write a new value into the located FFN to change a specific fact while leaving others intact [7].
- **Mass editing (MEMIT)**: scale editing to thousands of facts across layers [8].
- Caveat: edits can be brittle and leak—specificity and generalization trade off, so "localized" does not mean "cleanly isolated".

When parametric memory is not enough: retrieval augmentation

- Parametric storage is finite, stale, and hard to attribute.
- **RAG**: condition generation on documents fetched by a learned retriever, decoupling knowledge from parameters [10].
- **Retrieval during pretraining** (RETRO) lets a smaller model match larger ones by reading a trillion-token datastore [11].
- Benefits: updatable knowledge, source attribution, reduced hallucination on long-tail facts; cost: retriever quality and context-length pressure.

Failure modes: why stored knowledge misleads

- **Hallucination**: fluent output ungrounded in stored or retrieved facts; a surveyed, systemic property of generation [12].
- **Superposition interference**: packed features can fire spuriously, producing confident errors [4].
- **Stale parametric facts**: weights freeze the training snapshot; editing and retrieval are the two remedies above.
- **Attribution gap**: distributed storage makes it hard to say *why* a model answered—motivating interpretability tooling.

Synthesis and takeaways

- **Represent**: knowledge is directions in a superposed activation space, not discrete symbols [4, 5].
- **Store**: FFNs act as key–value memories; facts are distributed yet partly localizable and editable [6, 7].
- **Use**: attention routes information, FFNs recall it, the unembedding reads it out as logits [3, 9].
- **Engineer**: when parametric memory fails, retrieve (RAG/RETRO) or edit (ROME/MEMIT) [8, 10, 11].

References

[1] T. Mikolov, K. Chen, G. Corrado, and J. Dean. Efficient Estimation of Word Representations in Vector Space. *arXiv:1301.3781*, 2013.

[2] T. Mikolov, W. Yih, and G. Zweig. Linguistic Regularities in Continuous Space Word Representations. *NAACL-HLT*, 2013.

[3] N. Elhage et al. A Mathematical Framework for Transformer Circuits. *Anthropic / Transformer Circuits Thread*, 2021.

[4] N. Elhage et al. Toy Models of Superposition. *Transformer Circuits Thread*, 2022.

[5] T. Bricken et al. Towards Monosemanticity: Decomposing Language Models With Dictionary Learning. *Transformer Circuits Thread*, 2023.

References ii

[6] M. Geva, R. Schuster, J. Berant, and O. Levy. Transformer Feed-Forward Layers Are Key-Value Memories. *EMNLP*, 2021.

[7] K. Meng, D. Bau, A. Andonian, and Y. Belinkov. Locating and Editing Factual Associations in GPT. *NeurIPS*, 2022.

[8] K. Meng, A. Sharma, A. Andonian, Y. Belinkov, and D. Bau. Mass-Editing Memory in a Transformer. *ICLR*, 2023.

[9] C. Olsson et al. In-context Learning and Induction Heads. *Transformer Circuits Thread*, 2022.

[10] P. Lewis et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *NeurIPS*, 2020.

[11] S. Borgeaud et al. Improving Language Models by Retrieving from Trillions of Tokens. *ICML*, 2022.

[12] Z. Ji et al. Survey of Hallucination in Natural Language Generation. *ACM Computing Surveys*, 2023.